CUSTOMER DATA ANALYSIS

Insights Narrative Report

Tool: Microsoft Power BI

1. Executive Summary

This project uses Microsoft Power BI to explore customer shopping behaviour by gender, age group, product category and payment method. The dashboard reports 99.457K transactions, 299K products sold, total revenue of 251.51M, and an average customer age of 43.43.

2. Dashboard Overview

The dashboard image is placed here to give readers a visual overview before the detailed findings. It includes KPI cards, gender comparisons, product-category analysis, payment distribution and age-group charts.

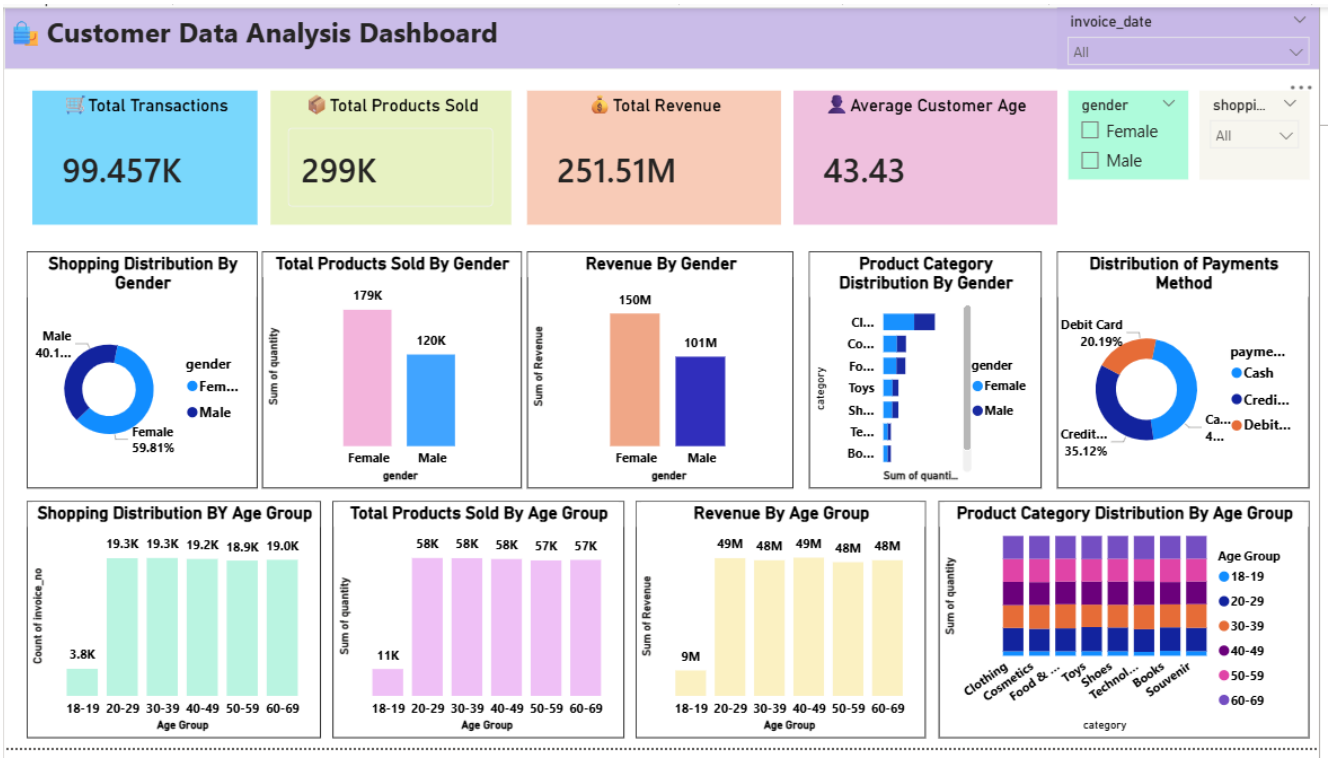


Figure 1. Customer Data Analysis Dashboard.

3. Gender Analysis

Female customers account for 59.81% of the shopping distribution shown. The dashboard reports approximately 179K products for female customers and 120K for male customers. Revenue is approximately 150M from female customers and 101M from male customers. In the displayed results, female customers contribute the larger share of shopping activity, product quantity and revenue.

4. Product Category Analysis

The category visuals compare quantities across product categories and show category performance by gender. Clothing is identified as the leading purchase category in the project findings. Some category labels are truncated in the screenshot, so exact category-level values should be checked in the interactive Power BI report.

5. Age Group Analysis

The dashboard compares age groups from 18–19 through 60–69. The 18–19 group has fewer transactions (about 3.8K), products sold (about 11K) and revenue (about 9M) than the other displayed age groups. The other groups are relatively close, with around 18.9K–19.3K transactions, 57K–58K products sold and 48M–49M revenue. This suggests the youngest group contributes less in the displayed dashboard.

6. Payment Method Analysis

The payment chart compares Cash, Credit Card and Debit Card. Credit Card is labelled 35.12% and Debit Card 20.19%; Cash appears as the largest segment and is identified in the project findings as the most commonly used payment method. The Cash percentage is not clearly legible in the screenshot, so it is not stated here.

7. Key Insights

- The dashboard summarises 99.457K transactions, 299K products sold and 251.51M in revenue.
- Female customers represent 59.81% of shopping distribution and contribute higher quantity and revenue in the displayed charts.
- Clothing is identified as the leading purchase category in the project findings.
- The 18–19 age group has lower transaction, quantity and revenue totals than the other displayed age groups.
- Cash is identified as the most commonly used payment method.

8. Recommendations

- Monitor popular categories such as Clothing and review category-level performance regularly.
- Use gender and age-group comparisons to plan targeted campaigns, based on observed data.
- Keep cash, credit card and debit card payment options convenient for customers.
- Use Power BI slicers for invoice date, gender and shopping mall to investigate selected segments.
- Validate measures and full category labels in Power BI before publishing exact values.

9. Conclusion

The Power BI dashboard provides a compact view of shopping behaviour across gender, age, product category and payment method. The displayed results show higher shopping and revenue contribution from female customers, lower activity in the 18–19 age group and useful differences in payment preferences. These insights can support better reporting and business decisions when combined with validation of the underlying data.

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